Joseph Mesches

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Electrical Engineering student at CSU and critical-care flight paramedic working at the intersection of clinical medicine and biomedical instrumentation. Now applying years of bedside clinical experience to machine-learning and explainable-AI systems for ECG interpretation. Focused on biomedical signal processing, clinical decision support, and embedded DSP.

Education

Bachelor of Science, Electrical Engineering — Expected May 2027
Colorado State University — Major GPA: 3.7 | Cumulative: 3.4 | Dean's List | HKN

Associate of Applied Science, Paramedicine — December 2019
Arapahoe Community College

Technical Skills

Programming: Python (NumPy, scikit-learn), C/C++, embedded C, MATLAB
Machine Learning & DSP: classification, explainable AI, feature extraction, adaptive filtering, FFT
Hardware & PCB: Altium Designer, KiCad, oscilloscopes, logic analyzers; SPI, I2C, UART
Biomedical Instrumentation: 12-lead ECG/EKG, EEG, analog biopotential front-ends

Clinical Experience

Critical Care / Flight Paramedic — 2017 – 2022
UCHealth LifeLine, Air Methods Corporation, Northglenn Ambulance
- Interpreted thousands of 12-lead ECGs (arrhythmia and STEMI recognition, defibrillation, pacing) alongside invasive hemodynamic, capnography, and co-oximetry monitoring for the most critically ill patients.
- Managed advanced mechanical ventilation, hemodynamic support, and critical-care procedures for trauma, ICU, and high-risk obstetric patients across scene, ED, and inter-ICU transport; interpreted blood gases and point-of-care labs.

EMS Field Training Officer, Instructor & QA Reviewer — 2015 – 2022
Multiple Colorado EMS Agencies
- Led field-training teams of up to 30 and trained 200+ paramedics; precepted students and developed field-training and continuing-education programs.
- Served on quality-assurance and education committees, reviewing clinical documentation and authoring SOPs to standardize care.

Research Experience

CUSP Research Intern — Summer 2025
Webb-Waring Institute, CU Anschutz
- Developed and executed an assay for oxidative-burst formation in immune cells; performed data analysis and interpretation of assay results.

Engineering Experience

Electrical Engineering Intern — June 2026 – Present
The RMH Group, Inc. (MEP)
- Reverse-engineered an automatic racking robot and conveyor system into comprehensive AutoCAD drawings; contributed to control-system projects improving efficiency at a toner-manufacturing plant.
- Maintained and updated as-built drawing packages (AutoCAD 2024, Adobe Acrobat Pro) for maintenance accuracy; invited to continue with the firm through the academic year.

Biomedical Engineering Projects

Explainable ECG Clinical Decision-Support System — Spring 2026
ECE 431 — Colorado State University
- Led integration on a 4-person team building a five-pipeline ML ensemble (XGBoost + 1D-ResNet CNN over engineered ECG signal-processing features) that classifies 12-lead ECGs into five diagnostic classes at 0.93 macro-AUC, on par with published deep-learning benchmarks.
- Authored the shared DSP library (filtering, R-peak detection, fiducial delineation) and made predictions auditable by mapping them to named clinical ECG criteria for clinician-reviewable decision support.

Rodent Activity Monitor — Fall 2025
CU Anschutz Neurotechnology Center
- Designed a custom ESP32 PCB (multi-sensor plus battery management; 5 deployed units, up to 14-day life) with a full-stack Docker/database/Grafana pipeline now in daily use by lab non-engineers.

Leadership

IEEE Student Branch — Vice President (2026–Present), President (2025–2026) — 2025 – Present
Colorado State University
- Lead branch operations, outreach, and communication between students, faculty, and industry partners.

Publications, Honors & Certifications

Publication: "Looking Past Dementia Reveals Hidden Life Threats," *Journal of Emergency Medical Services* (2018)
Certifications: Certified Flight Paramedic (FP-C, ICAPP), Critical Care Endorsement (P-CC), Nationally Registered Paramedic, Colorado EMS Primary Instructor
Honors: Walker-Weinshenker Family Scholar (2024–2026), Dean's List, IEEE-HKN, Compassionate Care Award (2018)